

The PLL (Phase Locked Loop) reference frequency is constantly compared to the disc's sync data (extracted from SRF) representing a frequency of 4.3218 MHz. When these two frequencies are equal the PLL is locked. If the PLL is not adjusted within its operating range it can cause incorrect disc rotation or dropout.

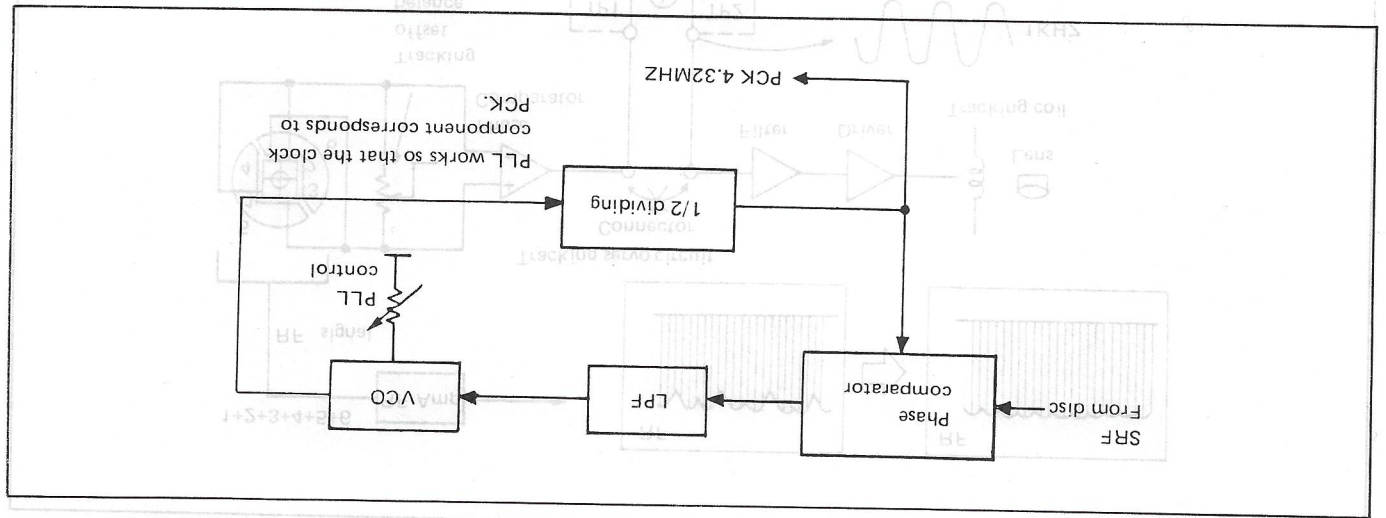


Figure 22-26

3. Various Defects of Disc & RF Waveform

Defects related to the disc are shown in figure 22-27 (A). Their respective RF waveform are shown in figure 22-27 (B).

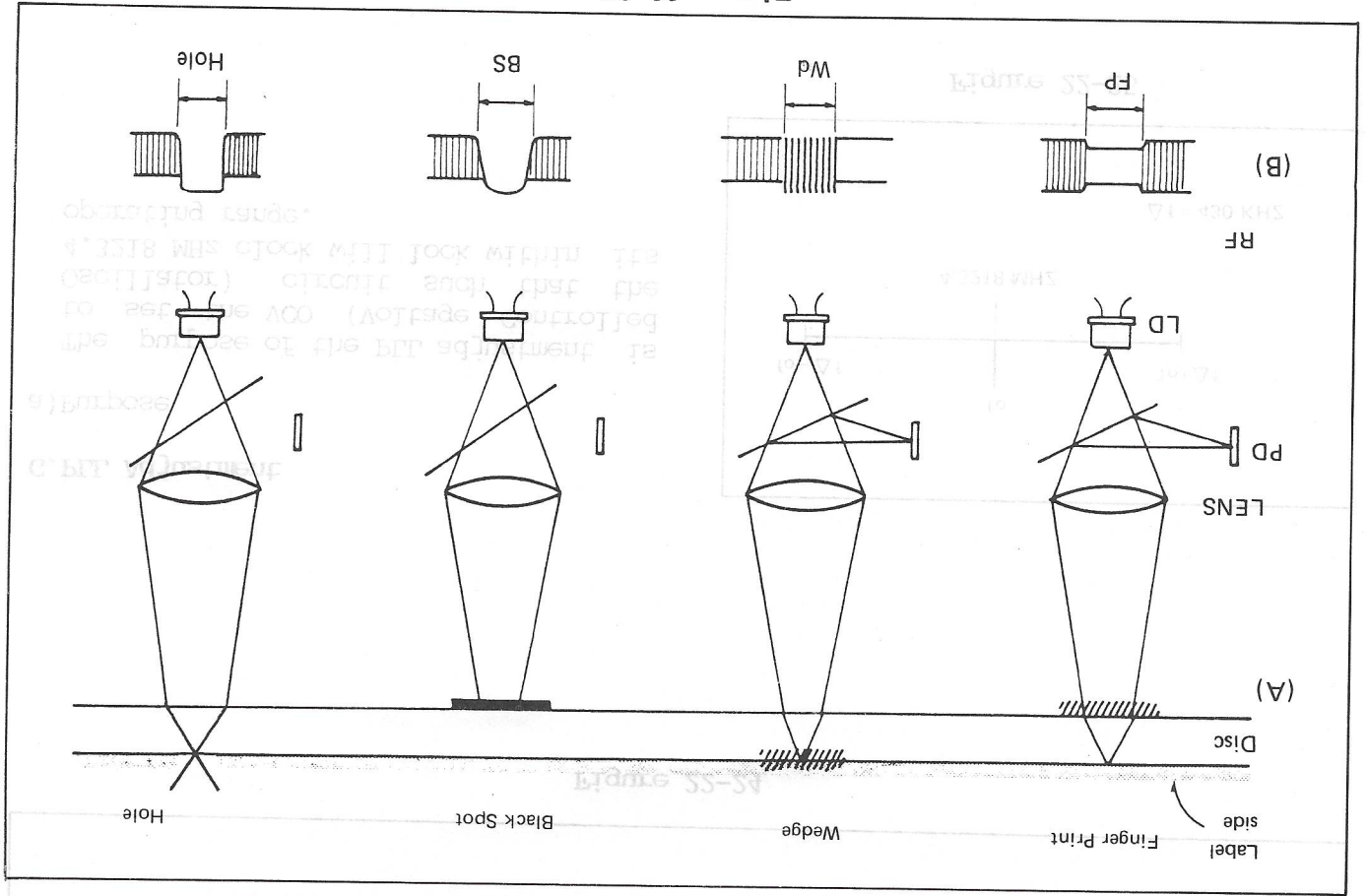


Figure 22-27

c) Adjustment method
The 1 KHz signal is fed through the tracking servo. The tracking offset balance control is to be adjusted for minimum and balanced level on the RF waveform.

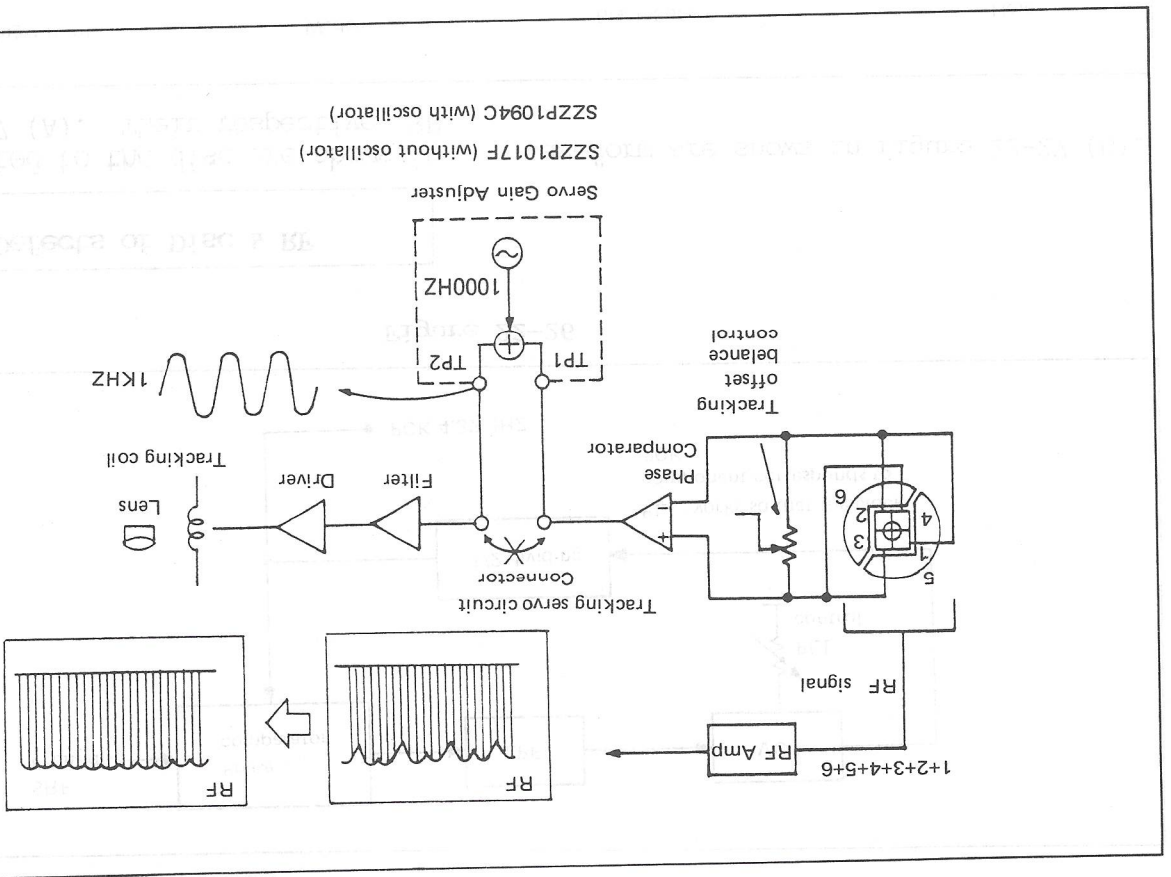


Figure 22-24

G. PLL Adjustment

a) Purpose

The purpose of the PLL adjustment is to set the VCO (Voltage Controlled Oscillator) circuit such that the 4.3218 MHz clock will lock within its operating range.

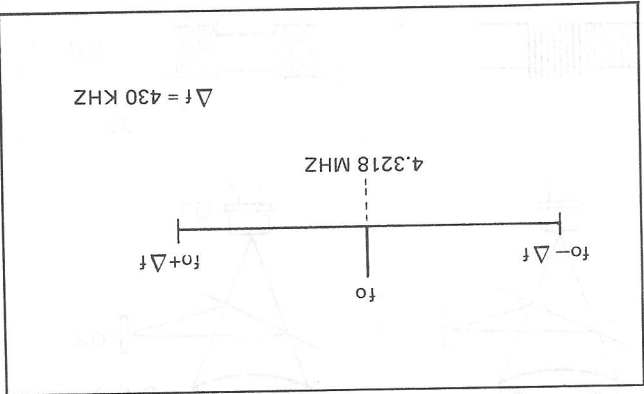


Figure 22-25

Information

We would like to inform you that we have developed new test fixture of **New Servo Gain Adjuster**. This new servo gain adjuster is improved version of former servo gain adjuster in order to adjust CD player quicker and accurately. To use this new servo gain adjuster, we believe CD player can be repaired much easier and repair time will decrease very much.

1. Main Feature

(1). Employed internal oscillator and power supply:

• It is not necessary to look for the power line on the main unit for connecting and prepare of external oscillator. The servo gain can be adjusted with just one connection of connector.

(2). Equipped servo gain judgment circuit:

• 3 LED indicate the condition of the servo gain and servo can be adjusted by monitoring those LED indication without connecting oscilloscope.

(3). Can be used any of Technics and Panasonic CD player:

• Including products introduced in the future, all of CD players are applicable to this new servo gain adjuster (except SL-P10, SL-P7 and SL-P8)

(4). Others:

- 8 kind of different frequency are available. and equipped test points for signal waveform monitoring.

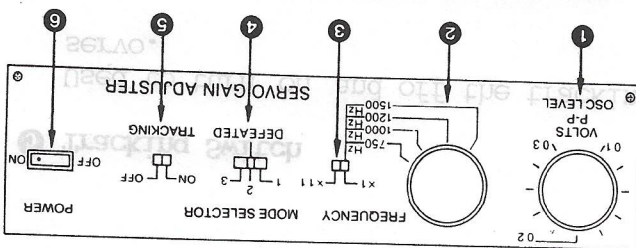
2. Name and Part Number

New Servo Gain Adjuster SZZP1094C

We are convinced that new test fixture is very useful and helpful for servicing of CD player and

3. Controls and Their Functions

Front View



① OSC Level Control

Used to adjust the output level of the built-in low frequency oscillator.

At minimum position (fully-counterclockwise position):

A 0.1 Vp-p level signal is delivered.

At center position:

A 0.2 Vp-p level signal is delivered.

At maximum position (fully-clockwise position):

A 0.3 Vp-p level signal is delivered.

② Frequency Control and Switch x1

Used to select the oscillation frequency.

Frequency	750Hz	1,000Hz	1,200Hz	1,500Hz		
Oscillation	x1	x1	x1	x1		
Frequency	x1.1	825Hz	1,000Hz	1,100Hz	1,320Hz	1,650Hz

We would like ask you to study to use it in your service network and place an order necessary quantity to us through normal spare parts order route. Furthermore, we would like to supply this new servo gain adjuster in future instead of former servo gain adjuster (SZZP1017F). and it will be discontinued after consumed all of our stock.

4. Servo Gain Adjustment

Procedures

1. Connect the 5-pin code supplied with this unit to the 5-pin connector on the rear panel.
2. Disconnect the servo adjustment connector on a CD player and connect the other end of the 5-pin code to the servo adjustment connector.
3. Set the modes selector switch (4) on the unit panel to position "2" (DEFEATED) and turn on the CD player to read the TOC.

Focus Servo Gain Adjustment

1. Adjust the input level for the CD player with the OSC level controller (1) on the unit panel. (Refer to the service manual for the input level).
2. Select the frequency of the oscillator with the frequency switches (2) and (3). (Refer to service manual for the frequency).
3. Set the mode selector switch (4) on the panel of the unit to position "3" and adjust the semi fixed volume for focus gain control on the CD player so that the LED "GOOD" on the servo gain INDICATOR panel lights up.

NOTE

First adjust the volume for focus gain control so that LOW LED on the servo gain INDICATOR panel lights up and then make further adjustments until the GOOD LED lights up.

4 Mode Selector Switch

Used to set the modes for tracking and focus servo adjustments.

1. Setting for tracking servo adjustment.
2. DEFEATED (through) for focus servo adjustment.
3. Setting for focus servo adjustment.

NOTE

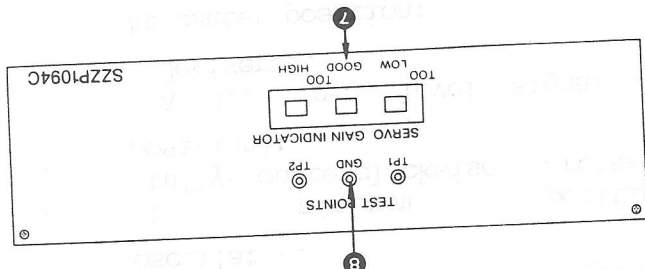
The adjustment settings (1 2 or 3) will depend on the model being used. Please refer to respective service manual.

5 Tracking Switch

Used to turn on and off the tracking servo.

6 Power Switch

Top View



7 Test Points

These are the in/output terminals for monitoring the servo amplifier. (Same I.P.s in the original servo gain adjuster).

8 Servo Gain Indicator

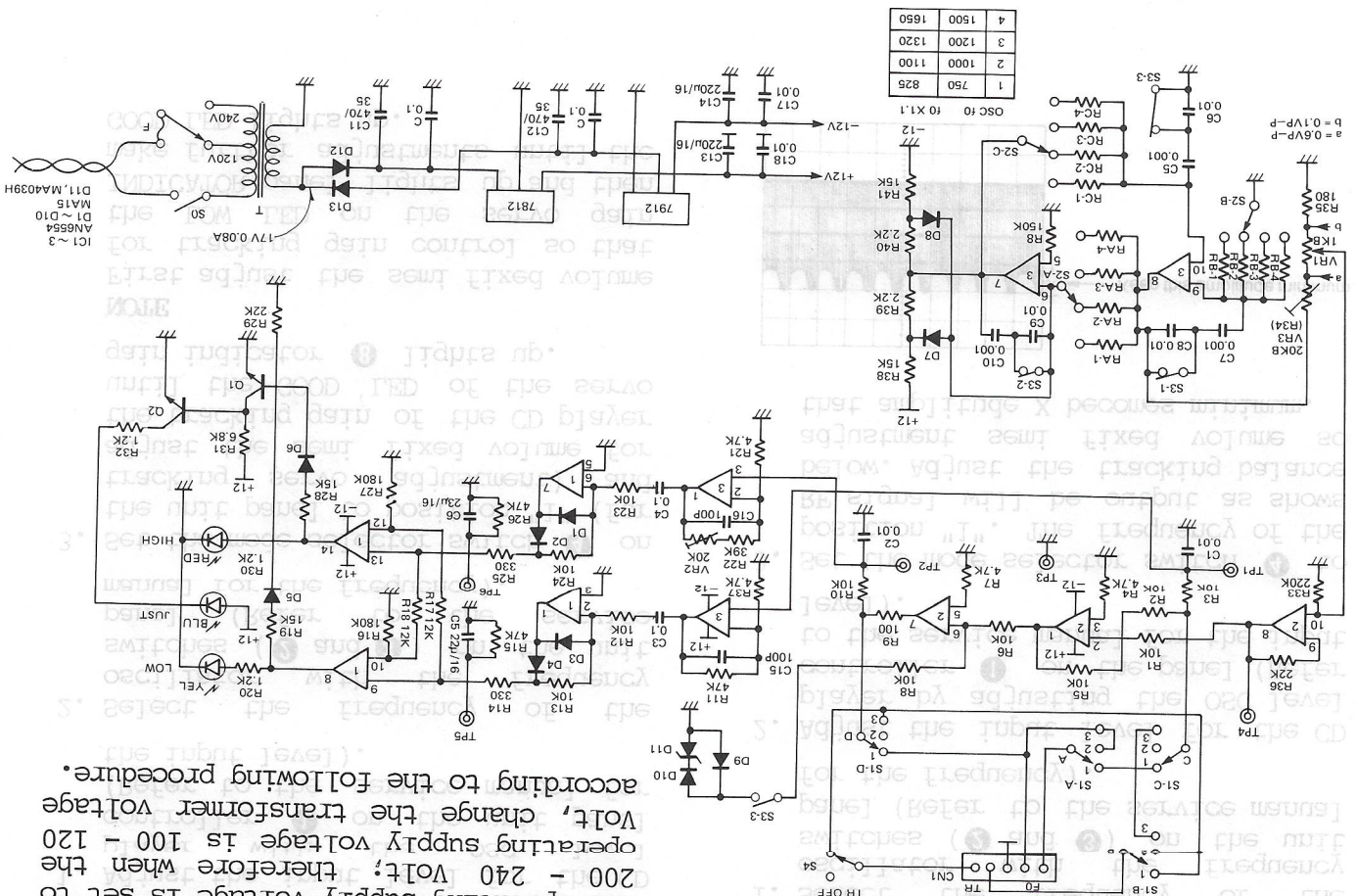
Displays the condition of the servo amplifier's gain during adjustment.

LED Status	Condition of the Servo Amplifier's Gain
LOW	Gain is low
GOOD	Gain is correct
HIGH	Gain is high

Schematic Diagram (SZZP1094C)

The operating supply voltage is set to 200 - 240 Volt; therefore when the operating supply voltage is 100 - 120 Volt, change the transformer voltage according to the following procedure.

CAUTION



Parts List

Parts No	Description	Q	Ref No	Parts No	Description	Q	Ref No
1SR35-200A	Diode	2	D12, D13	ERD25PJ223	Resistor	2	R36,R29
AN7812	IC	1	7812	ERD25PJ224	Resistor	1	R33
AN7912	IC	1	7912	ERD25PJ335	Resistor	1	R22
5P	Connector	1	CN1	ERD25PJ393	Resistor	1	R22
2SC1685Q	TR	2	Q102	ERD25PJ472	Resistor	1	R22
AN6554	IC	3	IC1 - IC3	ERD25PJ473	Resistor	4	R4,R7,R21
ECCF7H101J	C Capacitor	2	C15, C16	ERD25PJ682	Resistor	3	R11,R15
ECEA1CS220	Capacitor	2	C5, C6	ERD25PJ331	Resistor	1	R31
ECEA1CS221	Capacitor	2	C13, C14	ERD25CK105	Resistor	2	R14,R27
ECEA1VS471	Capacitor	2	C11, C12	ER025CK1183	Resistor	2	R14,RB4
ECQM1H103KZ	Polyster	4	C1,C2,C17	ER025CK1433	Resistor	1	RC4
ECQM1H104KZ	Polyster	4	C18	ER025CK1583	Resistor	1	RC3
ECQPH1H102FFZ	Polyster	3	C3,C4,C0,	ER025CK1743	Resistor	2	RA2 RB2
ECQPH1H103FFZ	Polyster	3	C5,C7,C10	ER025CK2103	Resistor	2	RC2
ERD25PJ101	Resistor	1	R9	ER025CK1333	Resistor	1	RC1
ERD25PJ103	Resistor	11	R1 -	EVHGHAF20-B13	Resistor	2	RA3 RB3
ERD25PJ122	Resistor	3	R20,R30	EVL70A24	VR	1	VR1
ERD25PJ123	Resistor	2	R17,R18	LN252RPH	VR	2	VR2,VR3
ERD25PJ153	Resistor	4	R19-	LN352GPH	LFD	1	H
ERD25PJ154	Resistor	1		LN452YPH	LFD	1	J
ERD25PJ181	Resistor	1	R35	MA150	LFD	1	L
ERD25PJ184	Resistor	2	R16,R27	MA4039	Diode	10	D1-D10
ERD25PJ222	Resistor	2	R39,R40			1	D11

Tracking Servo Adjustment

1. Adjust the input level for the CD player with the OSC level controller ① on the unit panel (Refer to the service manual for the input level).

2. Select the frequency of the oscillator with the frequency switches (② and ③) on the unit panel (Refer to the service manual for the frequency).

3. Set the mode selector switch ④ on the unit panel to position "1" (for tracking servo adjustment) and adjust the semi fixed volume for the tracking gain of the CD player until the GOOD LED of the servo gain indicator ⑧ lights up.

NOTE

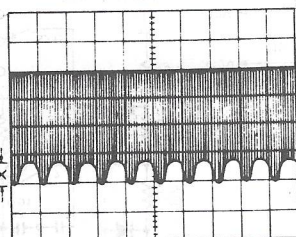
First adjust the semi fixed volume for tracking gain control so that the LOW LED on the servo gain INDICATOR panel lights up and then make further adjustments until the GOOD LED lights up.

Tracking Balance Adjustment

1. Select the frequency of the oscillator with the frequency switches (② and ③) on the unit panel (Refer to the service manual for the frequency).

2. Adjust the input level for the CD player by adjusting the OSC level controller ① on the panel (Refer to the service manual for the input level).

3. Set the mode selector switch ④ to position "1". The frequency of the RF signal will be output as shows below. Adjust the tracking balance adjustment semi fixed volume so that amplitude X becomes minimum.



Keep this amplitude minimum.

23. CD Changer Troubleshooting Flow Chart

